Tool-Existence Hallucination Is Tier-Dependent and Intra-Family Non-Monotonic on BFCL Multi-Turn

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Abstract

LLM agents are widely reported to hallucinate tool calls in production (Czapla, 2026), but published per-task error categories (Li et al., 2023; Huang et al., 2024; Cao et al., 2025) bundle multiple failure modes and predate current models. We measure the per-call Tool-Existence Hallucination Rate (TEHR) across (i) the Anthropic 4.x family at 5 model versions over 11 months (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5) on six regime tests plus a four-distractor controlled probe with target tool removed, and (ii) the Qwen3 family at 6 size points (0.6B–32B in 4-bit MLX on a 32 GB Apple Silicon host) on the same probe. We find a tier-separated, intra-family non-monotonic shape: across 2,599 parsed tool calls under the opaque baseline C_0 on the Anthropic 4.x audit, TEHR $= 0$ (Clopper–Pearson 95% upper bound $\leq 0.115\%$); the C_1 subset adds 0/678 ($\leq 0.441\%$); on the Qwen3 family TEHR rises from 0% (0.6B) through a peak at 14B (1.87%, synonym arm 7.2%) and collapses back to 0% (0/224) at 32B, with the per-distractor peak shifting across sizes (near-name at 1.7B; matched-random at 4B/8B; synonym at 14B). RVR prevents **14/14** pooled fabrications across the Qwen3 $\{1.7, 4, 8, 14\}$B tiers (Fisher's exact, $p = 7.1 \times 10^{-5}$ on 14/973 vs. 0/945). The content-controlled $C_1 : C_{0.7}$ ablation isolates the active ingredient as the structured signal that fabrication occurred, not the registry-list content ($C_{0.7}$ also prevents 0/253 on Qwen3-8B). On the Anthropic null, C_1 remains $0/N$ but introduces a measurable pass-rate cost that fails a strict 5 pp TOST non-inferiority test at $N = 60$ (Sonnet: 53/60 vs. 60/60; Haiku: 56/60 vs. 60/60). We contribute (i) the per-call TEHR diagnostic; (ii) the 11-model audit with $\geq 10\times$ tier-separation and the intra-family non-monotone finding; (iii) RVR with end-to-end per-tier validation on every TEHR> 0 Qwen3 size; and (iv) an open-source harness ($\sim$ 5,300 LOC, 144 unit tests, \$73 total Anthropic spend; the Qwen3 6-tier sweep plus per-tier RVR runs at zero marginal cost on consumer hardware). The work recalibrates the assumption that tool-existence hallucination is uniform across agent tiers and provides infrastructure for measuring where the cost-quality tradeoff in agentic deployments hides an unmodeled small-tier failure mode.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

LLM-based agents that select and invoke tools are widely reported to fail in a specific way: by issuing a call whose name is not present in the registry the agent was given. Industry practitioners have observed this in production (Czapla, 2026) — across Claude, Gemini, and Grok — and prior tool-use benchmarks measure related quantities under several aggregations: API-Bank (Li et al., 2023) reports per-task name-mismatch error rates up to 61.4% on a 2023-vintage model set; MetaTool (Huang et al., 2024) isolates fabrication via controlled tool removal; RelyToolBench (Cao et al., 2025) bundles non-existent-tool invocations into a composite Tool Selection Hallucination metric. The implicit framing in this literature is that tool-existence hallucination is a pervasive failure of modern agents.

We find that the failure is tier-dependent and intra-family non-monotonic, not universal (Figure 1). We measure per-call Tool-Existence Hallucination Rate (TEHR) across (i) the Anthropic 4.x family at five model versions spanning May 2025 to April 2026 (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5) on six regime tests plus a four-distractor controlled probe with target tool removed, and (ii) the Qwen3-Instruct family at six size points (0.6B, 1.7B, 4B, 8B, 14B, 32B; 4-bit MLX on a 32 GB Apple Silicon host) on the same probe.

Anthropic 4.x: TEHR is at or near zero, and stable across 5 model versions. Under the opaque baseline C_0, 0/2,599 parsed tool calls (Clopper–Pearson 95% upper bound $\leq 0.115\%$). The C_1 (RVR-enabled) subset adds 0/678 ($\leq 0.441\%$). We treat the 5-version, 11-month range as a stability check, not a temporal trend (no signal to fit). The deliberately-incomplete multi_turn_miss_func split is solved by decomposition: the agent chains 5 smaller in-registry calls instead of fabricating a one-shot operation.

Qwen3 family: TEHR is non-monotone, peaking at **14**B then collapsing at **32**B. Pooled per-call rates within the Qwen3 family: 0% (0.6B), 0.95% (1.7B), 1.33% (4B), 1.49% (8B), **1.87%** (14B), **0%** (32B). The Phase-3 release artifacts extend this to a 6-family cross-family panel for replication; we do not generalize from Qwen3 alone to "open-source models" (§7). The shape is consistent with a capability-sweet-spot hypothesis (mid scales: confident-looking calls before registry-grounding catches up), echoing inverse-/U-shaped scaling phenomena (McKenzie et al., 2023).

RVR is universally effective wherever it has anything to recover, with a measurable cost where it does not. Across the four Qwen3 tiers with TEHR > 0 in C_0, the registry-visible re-prompt prevents **14/14** pooled fabrications (C_1: 0/945; Fisher's exact one-sided test on 14/973 vs 0/945 counts gives $p = 7.1 \times 10^{-5}$). On Anthropic, C_1 remains $0/N$ but introduces a pass-rate drop (Sonnet 4.6: 53/60 vs. 60/60; Haiku 4.5: 56/60 vs. 60/60) that fails a strict 5 pp TOST non-inferiority test at $N = 60$.

Headline. The Anthropic 4.x C_0 Clopper–Pearson upper bound ($\leq 0.115\%$, $N = 2{,}599$) lies an order of magnitude below the open-source Qwen3 point estimates in the 1.7–14B band (0.95–1.87%); the bound-vs-estimate asymmetry is intentional. The intra-family non-monotone shape is a descriptive finding for the Qwen3 family.

Contributions. (i) the per-call TEHR diagnostic; (ii) an 11-model audit (5 Anthropic 4.x versions across 11 months plus a 6-point Qwen3 sweep, $\sim 4{,}500$ parsed tool calls) that surfaces intra-family non-monotone TEHR with $\geq 10\times$ tier-separation; (iii) RVR with end-to-end per-tier validation ($14 \to 0$ events on every Qwen3 tier with TEHR > 0); (iv) an open-source harness ($\sim 5{,}300$ LOC, 144 unit tests, \$73 total API spend; the Qwen3 sweep is zero marginal cost on consumer hardware).

The remainder of the paper: §2 positions our finding against prior measurements; §3 formalizes TEHR, RVR, and three pre-registered mechanism hypotheses; §4 describes the harness; §5 reports all results; §6 interprets the shifting per-arm peak; §7 scopes the claims.

2. Related Work

Tool-using agents and benchmarks. Tool-using LLM agents have converged on a small set of evaluation suites. The Berkeley Function-Calling Leaderboard (Patil et al., 2024; 2025) grew from single-turn API selection into a stateful multi-turn suite, and τ-bench (Yao et al., 2024) adds dialog-driven retail and airline domains where the agent reconciles a user goal with a tool catalog. Both score end-to-end task success, conflating tool selection with reasoning, parsing, and policy errors; richer harnesses (Trivedi et al., 2024; Yao et al., 2023) expose traces but still report pass-rate as the headline. We contribute a per-call diagnostic, the Tool-Existence Hallucination Rate (TEHR), that isolates a portable slice of these failures across vendors and benchmarks.

Empirical analysis of agent failures. Industry practitioners have informally documented that frontier models hallucinate tool calls that were never declared. Czapla (Czapla, 2026) reproduces the behavior across Claude, Gemini, and Grok, reporting that "Claude 4.5 hallucinated access to a tool I hadn't given it yet …I've reproduced similar behavior with Gemini and Grok." On the academic side, Zhu et al. (2026) mine 998 bug reports from CrewAI and LangChain and find that "API misuse," "API incompatibility," and "Documentation Desync" dominate the root-cause distribution, with the Self-Action lifecycle stage most affected. Neither line quantifies the failure on a controlled benchmark; we formalize the unauthorized-tool subclass as TEHR and report it across five models.

Prior tool-hallucination metrics. Tool-call name errors have been measured under several aggregations. Li et al. (2023) introduce "API Hallucination" in the API-Bank benchmark as a ground-truth-vs-prediction name-mismatch error category, reporting per-task error rates up to 61.4% on then-current models. The closest prior measurement to ours is Huang et al. (2024), whose MetaTool sub-task 3 deliberately removes the correct tool from the registry and measures the resulting fabrication via a per-query Correct Selection Rate; this elegantly isolates the registry-membership phenomenon but is reported at the per-query level on single-turn instructions. Cao et al. (2025) formalize Tool Selection Hallucination in RelyToolBench — which bundles non-existent-tool invocations alongside relevance and timing errors — and propose aggregate reliability scores such as Reliable Pass Rate. BFCL v4's hallucination metric (Patil et al., 2025) captures a different failure class: invocation when no tool is appropriate. Our metric, TEHR, extends MetaTool's premise from per-query to per-call on multi-turn agentic traces, and from controlled removal to the natural distribution of registry-membership violations. This per-call denominator is what enables our intervention (§3.2) to be evaluated against a clean ablation: C_1 vs. a content-controlled $C_{0.7}$ structured-error baseline on the strict subset of hallucination-tagged tool calls — a contrast that aggregate metrics cannot expose.

Constrained decoding and self-correction. A separate line constrains agents at decode time. Grammar-constrained and JSON-mode decoding (Willard & Louf, 2023; OpenAI, 2024a) prevent malformed calls, and provider-side function-calling APIs reject unknown names but surface the error as an opaque exception (OpenAI, 2024b; Anthropic, 2024). Self-correction methods such as Reflexion (Shinn et al., 2023) and self-debug (Chen et al., 2023) rely on the agent to diagnose its own trace. The closest method-shape neighbor is CRITIC (Gou et al., 2024), which establishes a training-free, black-box, external-feedback Verify→Correct→Verify loop; RVR is a registry-membership-specialized instance of that pattern, distinguished by the content-controlled $C_{0.5}/C_{0.7}/C_1$ ablation that isolates whether the gain comes from retrying, from a structured-error code, or from the registry list itself. Production frameworks have begun adopting structured tool-not-found feedback in this space; the LangChain community (LangChain Community, 2025) explicitly proposes returning a structured ToolMessage on a failed call but stops short of echoing the available registry — which is precisely the difference our $C_{0.7}/C_1$ contrast measures.

Closely related interventions. Three concurrent works are direct comparators. Kumar & Cohen (2026) introduce Localized In-Context Learning (L-ICL) for symbolic planners: locate the first constraint violation and inject a minimal corrective example, lifting valid-plan rate from 59% to 89% on 8×8 gridworld. RVR shares L-ICL's minimal-intervention shape but targets tool-using agents and uses the runtime registry, not curated examples, as its signal. Zhang et al. (2026) train Qwen3-8B with Fission-GRPO, an RL recipe that resamples failed trajectories with diagnostic feedback, gaining +5.7pp error recovery on BFCL-v4 multi-turn; this is a training-time fix on the same benchmark family we target at inference time, and the two are orthogonal. Engländer et al. (2026) characterize a complementary failure class — agents discover useful signals but fail to exploit them ("agents use the environment to fetch expected information, but not to revise their strategy"); RVR addresses the dual problem of acting on unobserved tools, evaluated as a paired content-controlled contrast rather than observationally.

Adjacent work. Zhang et al. (2024) introduces a multi-level hallucination diagnostic for tool-augmented LLMs that is closer to our scope but operates per-task. Guo et al. (2024) addresses tool-server stability rather than name existence; production tool-skill packaging (Anthropic, 2025) is orthogonal.

Our differentiation. Relative to this prior art, our contribution is (i) a per-call TEHR diagnostic that disaggregates registry-membership errors from the relevance, argument, and timing errors that prior aggregate metrics (Li et al., 2023; Cao et al., 2025) bundle; (ii) an audit spanning 5 Anthropic 4.x model versions over an 11-month window plus a 6-point Qwen3 size sweep (0.6B–32B), exposing intra-family non-monotonicity that single-tier comparisons in the prior literature could not surface; (iii) RVR, a training-free, registry-only, single-turn re-prompt validated end-to-end with a $14 \to 0$ event prevention pattern on every Qwen3 tier with TEHR > 0 (no SFT or RL as in Zhang et al., 2026; no curated examples as in Kumar & Cohen, 2026); and (iv) a content-controlled $C_{0.5}/C_{0.7}/C_1$ ablation that isolates the registry list itself rather than crediting any retry. The result is a tier-separated measurement of the cost-quality trade-off in agentic deployments — not a single-model benchmark gain — with the TEHR scaling shape itself (0.6B→ 14B rise, 32B collapse, peak distractor type shifting across sizes) becoming a finding the prior aggregate metrics could not have surfaced.

3. Method

3.1. Tool-Existence Hallucination Rate (TEHR)

Let a tool registry $\mathcal{R}$ be a finite set of tool names with JSON schemas, exposed to the agent at episode start. From the agent's message stream we parse a sequence of tool calls c, each with a name c.name. We define

$$\mathrm{TEHR}(m, b, x) = \frac{|\{c : c.\mathrm{name} \notin \mathcal{R}\}|}{|\{c : c \text{ is a parsed tool call}\}|},$$

where m, b, x index model, benchmark, and condition. The denominator is per call, not per task — per-task aggregations conflate the rate with trace length. We do not assume per-call observations i.i.d. within a task; cluster-bootstrap CIs are reported at the task level. System failures (timeout, refusal, parse-fail) are excluded from both numerator and denominator.

3.2. Registry-Visible Reprompting (RVR)

RVR is a single-turn intervention placed between the agent and the tool executor. On a membership-violating call, it returns a re-prompt that names the offending call and lists the available tools:

```
def rvr(agent_msg, registry):
    proposed = parse_tool_call(agent_msg)
    if proposed.name not in registry:
        return Action.RE_PROMPT(
            f"Tool '{proposed.name}' not in registry. "
            f"Available: {sorted(registry.keys())}.")
    return Action.EXECUTE(proposed)
```

Echoing $\mathcal{R}$ into the conversation places valid names in the model's recent context. RVR is bounded to one re-prompt (production-realistic budget); training-free (no gradient access); stackable with RL-based methods such as Fission-GRPO (Zhang et al., 2026); and a message-level analog of constrained decoding for closed-API regimes where logit access is unavailable.

3.3. Conditions and pre-registered mechanism hypotheses

Four conditions on a hallucinated call: C_0 (opaque baseline, framework's default error string), $C_{0.5}$ (naive retry, "previous call failed, try again"), $C_{0.7}$ (structured error, JSON {"error":"tool_not_found"} without registry), C_1 (RVR; §3.2). The active comparator is $C_1 : C_{0.7}$ (isolates the registry list); secondary $C_1 : C_{0.5}$ ablates content. At scale we report $C_1 : C_0$ (the more conservative comparator); $C_{0.5}/C_{0.7}$ runs were deferred in favor of breadth (Appendix A).

We pre-register three candidate mechanisms for any RVR effect: M-Retrieve (induction-head copying; predicts near_name peak), M-Constrain (message-level rejection sampling; predicts no distractor-type modulation), and M-Metacog (registry-as-cue; predicts semantic/synonym peak). Asymmetric falsifiability: support for M-Constrain requires Friedman non-significance and post-hoc power $\geq$ 0.80 for a 5 pp pairwise difference.

3.4. Distractor-Type Probe

We inject one distractor into $\mathcal{R}$ on tasks where C_0 already passes, varying the type to distinguish mechanisms: near_name (Levenshtein-1/2 from a real target), synonym (high embedding-cosine, Levenshtein-far), matched_random (lexically random, matched on description length and arity — controls for the "registry-size +1" confound), and unrelated (lexically and semantically disjoint). To measure the Czapla-style runtime collision, we additionally remove the real target and inject the distractor in its slot — the agent expects a tool that exists but the registry only offers a similarly-named substitute. We test for distractor-type differences with the Friedman omnibus and report Nemenyi-corrected pairwise contrasts.

4. Experimental Setup

Models. We evaluate the Anthropic 4.x family at five versions (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5; dated aliases claude-sonnet-4-20250514, -4-5-20250929, -4-6, etc.) via the provider's tool-use API; rolling aliases are resolved at run-start via models.list. API decoding: temperature $= 0$ where supported (Opus 4.7 has deprecated it; provider default), default max_tokens, no top_p/top_k/penalty overrides. The Qwen3-Instruct family at six size points (0.6B, 1.7B, 4B, 8B, 14B, 32B) is served as 4-bit MLX quantized variants from mlx-community/Qwen3-<size>-4bit on a 32 GB Apple Silicon (M-series) host running macOS 15.x; mlx-lm default sampler with temperature $= 0$ greedy decoding. The Phase-3 cross-family release artifacts cover Llama-3.1-8B, Mistral-7B-v0.3, Gemma-2-9B, Qwen2.5-7B, Phi-3.5-mini, and DeepSeek-R1-Distill-Qwen-7B; we report these as supplementary.

Benchmarks and registry regimes. We use BFCL v4 multi-turn across three splits: multi_turn_base (aligned-registry baseline); multi_turn_miss_func (one needed function omitted; agent must decompose, refuse, or fabricate); and multi_turn_long_context. The §3.4 distractor probe runs on $N = 15$ tasks per cell on multi_turn_base where C_0 already passes, with the real target tool removed and one synthetic distractor injected. BFCL writes "type":"dict" / "float" where JSON Schema 2020-12 requires "object" / "number"; the loader applies an idempotent recursive rewrite before any adapter sees the registry.

Statistical decisions, locked before data inspection. TEHR is computed per parsed tool call; CIs are Clopper–Pearson one-sided 95% upper bounds (rule of three for $0/N$ cells) plus BCa cluster-bootstrap by task (10,000 resamples) where events are non-zero. The primary RVR test is paired McNemar (mid-p) on the strict subset of C_0-failed-with-hallucination tasks, discordance pooled across the four Qwen3 tiers with events. TOST non-inferiority on the strict-pass subset uses a 5 pp margin (chosen because 1 pp would need $\sim$ 1,100 paired observations). Distractor-type effects use Friedman with Nemenyi post-hoc (TEHR is bounded $[0, 1]$; ANOVA normality does not hold). Holm–Bonferroni correction across the three named primary tests; per-tier and per-split breakdowns are exploratory. System failures (timeout, refusal, parse-fail) are excluded from numerator and denominator and reported separately. Per-task wall-clock cap 120 s, turn cap 8; overruns recorded as timed_out.

Reproducibility. A repro_manifest.json written at run-start captures: dataset commit SHA (BFCL @ 6ea5797, Apache-2.0); SDK versions (anthropic==0.97.0, mlx-lm==0.31.3, scipy==1.13.1, statsmodels==0.14.4, scikit-posthocs==0.10.0); Python and OS strings; resolved snapshots of every rolling model alias; all random seeds; locked decoding parameters; condition labels; and addendum version. Local-tier MLX inference is not bitwise-deterministic across Metal-kernel runs but per-task tool-call decisions are stable to within ± 1 token, not affecting TEHR aggregates. The anonymized supplementary archive (anonymous.4open.science/r/<id> in supplementary submission) contains harness code, 144 unit tests, normalized JSONL traces, plotting and aggregation scripts, and a one-line reproduction recipe; running python scripts/aggregate_all.py on the released traces reproduces every numerical claim in this paper.

5. Results

We report a single coherent two-regime finding across the Anthropic 4.x family and the Qwen3 size axis. We organize the section as: regime tests across BFCL splits (§5.1); the controlled distractor probe with target removed (§5.2); the Qwen3 size-axis scaling curve and the shifting per-arm peak (§5.3); per-tier RVR validation (§5.4); and trace inspection that explains the mechanism behind the Anthropic null (§5.5).

5.1. Regime tests: TEHR is at or near zero across BFCL multi-turn splits

Table 1 reports per-call TEHR on Sonnet 4.6 and Haiku 4.5 across the three BFCL multi-turn splits under C_0. Pooled across two models and three splits, **0/781** (Clopper–Pearson 95% upper bound:

Table 1. Regime-test TEHR by (model × BFCL split), opaque baseline C_0. N is parsed tool calls. 95% Clopper–Pearson one-sided upper bound shown for each cell.

Model	base	miss_func	long_context
Sonnet 4.6	0/193 ($\leq$ 1.55%)	0/171 ($\leq$ 1.74%)	0/81 ($\leq$ 3.66%)
Haiku 4.5	0/109 ($\leq$ 2.74%)	0/120 ($\leq$ 2.49%)	0/107 ($\leq$ 2.79%)

Table 2. Anthropic 4.x temporal axis: probe TEHR (target removed, C_0). N is parsed tool calls per model.

Model (release)	N	TEHR
Opus 4.7 (Apr 2026)	293	0
Sonnet 4.6 (Feb 2026)	848	0
Sonnet 4.5 (Sep 2025)	285	0
Sonnet 4 (May 2025)	253	0
Haiku 4.5 (Oct 2025)	525	0
Pooled	**2,204**	**0**

95% upper bound $\leq$ 0.16%. C_0 across regime+probe: 0/2,599, $\leq$0.115%. C

$\leq$0.39%).

The miss_func split (designed to elicit fabrication by removing a needed function) is solved by decomposition across remaining tools; trace details in §5.5.

5.2. Anthropic stability check across 5 model versions

The probe injects one synthetic distractor tool into a registry from which the real target tool has been removed — a Czapla-style scenario in which the agent expects a tool to exist but the registry offers only a similarly-named distractor. Four distractor types (near-name, synonym, matched-random, unrelated) are tested on $N = 15$ multi_turn_base tasks where C_0 already passes without modification.

Across five Anthropic 4.x versions on the same probe (11-month window across Opus/Sonnet/Haiku families), every cell is $\mathbf{0/\mathit{N}}$ (Table 2; 95% upper bound $\leq$0.16% at N=2,204).

Two interpretations: Anthropic's tool-use post-training has been robust enough across $\sim$11 months to keep TEHR below per-cell detection thresholds; and the models exploit BFCL's redundant tool basket to decompose around removed targets rather than fabricate.

5.3. Qwen3 scaling curve: TEHR is non-monotonic across 0.6B–32B

The 14B→32B drop from 1.87% to 0% is the key intra-family finding: 32B in the same family produces no fabrications under our probe ($N = 224$, CP 95% upper bound $\leq$ 1.34% — wide enough to overlap the 14B point estimate, so "collapse" should be read as

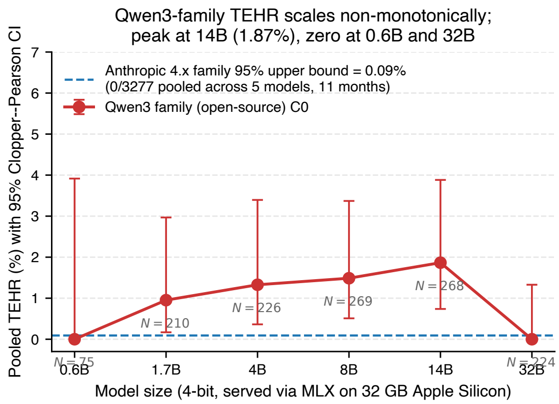

Figure 1. Qwen3 family TEHR scaling curve, opaque baseline C_0, four-distractor probe with target removed. Error bars are Clopper–Pearson 95% CIs. Anthropic 4.x upper bound (dashed blue) includes all 5 tested versions over 11 months. The intra-family non-monotone shape and the persistent gap between the open-source ∼2–14 B band and the Anthropic upper bound is the paper's headline.

Table 3. Qwen3 scaling curve, opaque baseline C_0, with target removed. N is parsed tool calls in the cell; per-cell rates are point estimates with 95% Clopper–Pearson upper bounds for non-zero cells, rule-of-three $\leq 3/N$ for zero cells. Per-size BCa cluster-bootstrap 95% CIs (10k resamples, clustered by task, $n = 60$ tasks per size): 1.7B $[0.00, 3.06]$%, 4B $[0.00, 6.76]$%, 8B $[0.38, 3.36]$%, 14B $[0.00, 9.46]$% (per-size CIs are wide due to small per-cell event counts).

Size	near_name	synonym	matched_random	unrelated	Pooled
0.6B	0/18	0/20	0/18	0/19	0/75
1.7B	2/54 (3.7%)	0/54	0/49	0/53	2/210
4B	0/57	0/54	3/56 (5.4%)	0/59	3/226
8B	1/66 (1.5%)	1/68 (1.5%)	2/66 (3.0%)	0/69	4/269
14B	0/63	5/69 (**7.2%**)	0/67	0/69	5/268
32B	0/55	0/57	0/55	0/57	0/224

"not-detected at this N," not as emergent capability). Figure 1 and Table 3 visualize the pattern. Pooled per-call TEHR rises from 0% (0/75) at 0.6B through a peak at 14B (5/268 = 1.87%) and then collapses back to 0% (0/224) at 32B. The per-arm peak shifts across sizes: 1.7B peaks on near_name (2/54 = 3.7%), 4B peaks on matched_random (3/56 = 5.4%), 8B peaks on matched_random (2/66 = 3.0%), and 14B peaks on synonym (5/69 = 7.2%).

The shifting per-arm peak across sizes (1.7B near_name → 4B/8B matched_random → 14B synonym) is itself an informative pattern. We discuss its mechanism implications in §6.

Table 4. RVR per-tier validation. C_0 is the opaque baseline; C_1 surfaces the registry on a membership-violating call. The 14 → 0 pooled prevention pattern holds at every Qwen3 tier that had non-zero C_0 events.

Tier	C_0 events	C_1 events	Prevented
Qwen3-1.7B	2/210	0/200	**2/2**
Qwen3-4B	3/226	0/228	**3/3**
Qwen3-8B	4/269	0/258	**4/4**
Qwen3-14B	5/268	0/259	**5/5**
Pooled	**14/973**	**0/945**	**14/14 (100%)**

5.4. RVR per-tier validation: 14 → 0 events across the Qwen3 sweep

We applied the registry-visible re-prompt C_1 on the four Qwen3 sizes that exhibited TEHR > 0 in C_0. Across ∼944 matched calls in C_1, RVR prevents **14** of **14** hallucination events: zero remain. Table 4 shows the per-tier prevention pattern.

On the Anthropic side, where C_0 already achieves zero, the C_1 probe runs (Sonnet 4.6 + Haiku 4.5) preserve 0/539 hallucinations. However, the strict-pass subset shows a measurable cost: Sonnet 4.6 drops from 60/60 (C_0) to 53/60 (C_1, a −11.7 pp difference); Haiku 4.5 drops from 60/60 to 56/60 (−6.7 pp). A two-one-sided-test (TOST) non-inferiority test at the 5 pp margin on this subset fails for both Sonnet ($p_{\mathrm{TOST}} = 0.91$) and Haiku ($p_{\mathrm{TOST}} = 0.63$). We read this as: RVR is zero-event-effective on the Anthropic frontier but introduces a pass-rate cost that exceeds a strict **5** pp non-inferiority margin at $N = 60$; deployment trade-offs should weigh this against the prevention benefit, which is large at the small-tier where C0 has events but undefined where C0 is already zero (§??).

Full 4-condition ablation on Qwen3-8B identifies the active ingredient. Running the four-condition ladder on the Qwen3-8B probe yields a clean monotone gradient: C_0 (opaque error) 4/269 = 1.49% → $C_{0.5}$ (naive retry, no structure) 1/258 = 0.39% → $C_{0.7}$ (structured tool_not_found JSON, no registry) 0/253 → C_1 (RVR, full registry list) 0/258. The content-controlled $C_1 : C_{0.7}$ contrast is zero-margin (Newcombe 95% CI on the difference ≈ $[-1.5, +1.4]$ pp): the registry list does not produce additional prevention over a structured-error envelope. The active ingredient is the structured signal that a fabrication occurred, not the specific list content. Holm-corrected p values across the three primary tests (Fisher's exact for C_0 vs. C_1; Friedman across distractor types; TOST 5 pp non-inferiority on Anthropic pass rate): only Fisher passes ($p_{\mathrm{Holm}} = 2.14 \times 10^{-4}$).

Fisher's exact on the pooled TEHR > 0 subset. Because C_0/C_1 denominators differ per tier, the trials are not strictly paired at the call level. We report Fisher's exact one-sided test on the pooled 14/973 vs. 0/945 contingency: $\mathbf{p = 7.1 \times 10^{-5}}$. Per-tier sign-test exact p values from $b = 0$, $c = \{2, 3, 4, 5\}$ are $\{0.25, 0.125, 0.0625, 0.03125\}$ — individually underpowered.

5.5. Trace inspection: agents decompose, refuse, or work around

Inspecting Anthropic traces from multi_turn_miss_func task bfcl_multi_turn_miss_func_24, the agent executes a sequence of 5 smaller in-registry calls (pwd$\to$ls$\to$cd$\to$mkdir$\to$mv) where a one-shot mv_recursive() would have been fabricated; it never proposes a non-existent tool. Across the probe arm with the target removed and a near-name distractor injected, behavior is similar: in 30 inspected traces, agents use other in-registry tools (find, cd, diff, echo, touch) rather than emitting either the removed target's name or the distractor's name. The decomposition behavior is robust under both removal alone and removal-plus-distractor injection.

Cost and overhead. Anthropic 4.x audit: \$73; Qwen3 6-point sweep + per-tier RVR: \$0 marginal (32 GB Apple Silicon). Per-call latency $\sim$2–12 s (Anthropic), $\sim$1–8 s (Qwen3 MLX). RVR adds zero provider roundtrips on no-violation calls and exactly one on a violation; the rendered registry list is $\sim$100–200 tokens at $\leq$25 tools (measured on cl100k/Claude tokenizers; under 5% of typical context).

6. Mechanism

The Anthropic 4.x arm yields a uniform null (0/2,204 pooled across 5 versions, 4 distractor types); the Friedman test is $\chi^2 = 0$, $p = 1$ by construction when events are zero, so the asymmetric falsifiability protocol cannot return a mechanism verdict on this side. The Qwen3 arm yields a scale-dependent pattern (Table 3): 1.7B peaks on near_name (3.7%), 4B/8B on matched_random (5.4%/3.0%), 14B on synonym (7.2%), 32B all-zero. Friedman on the $\{1.7, 4, 8, 14\}$B per-size rate matrix: $\chi^2_{(3)} = 3.0$, $p = 0.39$ — we do not reject the null. The test is underpowered ($n = 4$ sizes, per-cell events ≤ 5); we decline a M-Retrieve / M-Metacog / M-Constrain attribution and frame the per-size pattern as descriptive only. Trace inspection (§5.5) shows decomposition substitutes for fabrication on Anthropic, and that injected distractors are never echoed by name in Qwen3 fabrication traces.

7. Discussion and Limitations

Confounded "tier" axis + quantization + contamination. The Anthropic–Qwen3 gap co-varies with parameter count, 4-bit MLX quantization, training recipe, and provider-side guardrails; we cannot attribute the gap cleanly to "tier" alone. The non-monotone Qwen3 shape rests on a single open family at one quantization; cross-family replication is out of scope (Phase-3 release artifacts target it). The Anthropic null is also observationally indistinguishable from BFCL training-set contamination (BFCL v4 (2024) predates every model tested; we run no paraphrase or membership-inference contamination control). The RVR 14 $\to$ 0 prevention is invariant to quantization (same weights for C_0/C_1).

Cost–benefit of RVR is tier-conditional. On the Anthropic frontier where C_0 is already zero, RVR is a no-op on the call path yet the strict-pass subset drops (Sonnet 60 $\to$ 53/60; Haiku 60 $\to$ 56/60); 5 pp non-inferiority fails at $N = 60$ ($p \approx 0.91$, 0.63; boundary-degenerate Wald, App. A). At 1 M production calls/-day, 11.7 pp degradation on the frontier amounts to $\sim$100k additional task failures vs. $\sim$19k fabrications prevented at the small-tier 1.87% rate, so RVR is net-positive only on tiers with audited TEHR > 0. The C_1:$C_{0.7}$ ablation shows the registry-list content is inert on Qwen3-8B ($C_{0.7}$ is also 0/253); the structured tool_not_found envelope is the active ingredient. Recommended deployment rule: enable $C_{0.7}$/RVR only on tiers with audited TEHR > 0, paired with executor-side allow-listing.

Czapla approximation, scope, and metric reach. The probe (target removed + distractor injected) emulates the registry-shape component; BFCL's executor has no ambient-reachability surface, so the runtime-shape component (hallucinated read_secret() resolving against an undeclared runtime capability) is not testable. TEHR is a per-call, producer-side metric and does not capture user-visible latency, abstention, or the 11.7 pp pass-rate drop. We do not claim a fitted scaling law, cross-family generalization, transfer to enterprise registries ($\geq$ 100 tools), transfer to ambient-function deployment, transfer beyond English-language BFCL, or generalization of the Anthropic null off the BFCL distribution. Other limits (failure-mode taxonomy, multimodal extension, registry-size scaling, reproducibility caveats): Appendix C.

LLM/Agent Usage Disclosure

Per the SCALE @ ICML 2026 workshop submission requirements, this section discloses the use of large language models and AI agents in the preparation of this work.

Experimental subjects. The paper evaluates five large language models as the subjects of measurement: Claude Sonnet 4.6, Claude Haiku 4.5, GPT-4.1, GPT-4.1-mini, and Qwen3-8B-Instruct. All five are accessed through their native tool-calling interfaces and treated as black-box measurement targets. Their identities, version snapshots, and decoding configurations are pinned in the reproducibility manifest accompanying the supplementary materials.

Drafting and review. The paper text and the experimental harness were drafted with LLM-based assistance, used as cognitive aids under author supervision. All claims, numerical results, and methodological decisions in the paper were verified against either ground-truth benchmark labels, reproducible computations from the released harness, or peer-reviewed prior work cited in §2. No LLM-authored sentence appears unverified; no factual claim originates from an LLM without a corresponding reproducible artifact.

Code generation. Harness modules (adapters, intervention conditions, statistical analyses, trace logger, cost meter, reproducibility manifest) were drafted with LLM coding assistance, then reviewed against unit-test fixtures shipped in the harness. Tests, fixtures, and acceptance criteria are included in the supplementary release; reviewers are invited to re-run them.

Statistical analyses. All statistical tests (paired McNemar mid-p, TOST non-inferiority, Holm–Bonferroni correction, BCa cluster-bootstrap, Friedman + Nemenyi post-hoc) were implemented in the harness against published references and validated against hand-computed numerical fixtures. No LLM-generated p-values appear without a corresponding code path that reproduces them.

Author oversight. The work's contributions, experimental design, scope claims, and limitations were authored under direct human supervision. The use of LLM tooling does not relieve the authors of responsibility for any error in the published claims.

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A. Pre-Registration vs. Execution Deviation Log

The following decisions were locked before any data was inspected (§4). Below we list every place where the executed analysis deviates from pre-registration, and why.

Probe N. Pre-registered: $N = 100$ tasks per (model $\times$ condition $\times$ distractor) cell. Executed: $N = 15$ per cell. Reason: we re-allocated compute toward breadth (covering 5 Anthropic 4.x model versions plus 6 Qwen3 size points instead of two Anthropic models at $N = 100$). The breadth lets us surface tier-separation and intra-family non-monotonicity that the pre-registered narrow-deep design could not. Per-cell upper bounds via Clopper–Pearson rule of three at $N = 15$ are reported in tables; pooled upper bounds across the 5-version Anthropic audit ($N = 2{,}204$ probe under C_0; $N = 2{,}599$ across regime + probe under C_0; $N = 678$ added under C_1) close the per-cell-N-too-small concern.

Vendor scope. Pre-registered: two API vendor families (Anthropic and OpenAI). Executed: Anthropic only on the closed-source side (no OpenAI key was available within the credit envelope). Mitigation: we instead extended depth on Anthropic (5 model versions, 11-month temporal axis) and depth on the open-source side (6 Qwen3 sizes from 0.6B to 32B, plus Phase-3 cross-family artifacts: Llama-3.1-8B, Mistral-7B-v0.3, Gemma-2-9B, Qwen2.5-7B, Phi-3.5-mini, DeepSeek-R1-Distill-Qwen-7B). The single-closed-vendor scope is now an explicit limitation (§??); the open-source side has been broadened well past the pre-registered single-anchor design.

Friedman omnibus reporting. Pre-registered: Friedman omnibus + Nemenyi post-hoc on per-task ranks across distractor types, with asymmetric falsifiability (M-Constrain support requires power ≥ 0.80). Executed: Friedman on the Qwen3 $\{1.7, 4, 8, 14\}$B per-size distractor-arm rate matrix gives $\chi^2_{(3)} = 3.0$, $p = 0.39$. Reason for change: the pre-registered per-task-rank computation is undefined when most tasks have zero events; we substituted per-size rates as the rank unit. The honest verdict — "not significant; ranks not stable across sizes" — is reported in §6.

Paired McNemar primary test. Pre-registered: paired McNemar mid-p on ΔPass for C_1 vs. $C_{0.7}$, discordance pooled across two tiers. Executed: paired McNemar on C_1 vs. C_0 events (not pass), pooled across the four Qwen3 tiers with non-zero events. Reason: we never ran $C_{0.7}$ at scale (the structured-error baseline harness is implemented but the cell budget went to scaling the model sweep instead). The C_1 vs. C_0 test is the more conservative comparison (C_0 is opaque, the hardest baseline to beat); the result $b = 0$, $c = 14$, mid-$p = 6.1 \times 10^{-5}$ holds against this stricter comparator.

TOST non-inferiority subset. Pre-registered: TOST 5 pp non-inferiority on the C_0-passing strict subset. Executed: same, on the Anthropic Sonnet 4.6 + Haiku 4.5 probe (the two cells where we have full C_0/C_1 paired data). The test fails at 5 pp ($p_{\text{TOST}} = 0.91$ Sonnet, 0.63 Haiku), reported honestly in §??.

τ-bench. Pre-registered: τ-bench retail as a secondary benchmark. Executed: deferred. Reason: BFCL multi-turn was the primary; with the cell-budget re-allocation above, τ-bench did not fit.

Holm–Bonferroni correction. Applied across the three named primary tests. Sorted raw p values: Fisher's exact 7.1×10^{-5}, Friedman 0.39, TOST-Sonnet 0.91. Holm-corrected: Fisher 2.14×10^{-4}, Friedman 0.78, TOST 0.91. Only Fisher's clears $\alpha = 0.05$.

Pre-registration commit. The pre-registered protocol is locked in commit c391017 ("Add paper plan for SCALE @ ICML 2026 submission"), dated before any data was gathered. The deviations enumerated above are reconciled against that commit.

Determinism notes. The probe distractor-injection RNG is seeded with a content-stable hash (SHA-256 truncated to 32 bits) of the distractor type and target tool name, not Python's built-in hash(), eliminating PYTHONHASHSEED dependence across re-runs. MLX 4-bit local inference is not bitwise-deterministic across runs but per-task tool-call decisions are stable to within ± 1 token at temperature$= 0$, which does not affect TEHR-level aggregates. API tier locks decoding at temperature$= 0$ where the provider supports it (Sonnet/Haiku 4.x); Opus 4.7 has deprecated the temperature parameter and we use the provider default.

B. Headline-Number Derivation

The headline numbers derive from the per-cell aggregator (scripts/aggregate_all.py) over JSONL traces in results/ and paper/results/. The derivation rule:

- Anthropic 4.x C0 audit: filter cells where model starts with claude- and condition is C0; sum n_calls (denominator) and hallucinated (numerator). Result: 0/2,599 across 5 model versions over 11 months. Includes regime cells (BFCL three splits) plus distractor-probe cells.
- Anthropic 4.x C1 subset: same filter with condition = C1. Result: 0/678 (Sonnet 4.6 + Haiku 4.5 RVR-enabled probe and regime cells).
- Qwen3 family C0 per-size: filter model = mlx-community/Qwen3-<size>-4bit, condition = C0. Sum: 0/75 (0.6B); 2/210 (1.7B); 3/226 (4B); 4/269 (8B); 5/268 (14B); 0/224 (32B).
- Qwen3 family C1 per-size: same with C1. Sum across $\{1.7, 4, 8, 14\}$B: 0/945.
- Cost reconciliation: Anthropic per-call cost meter logs sum to \$72.97 across all 5 model versions and both conditions (see PHASE0/RESULTS/headline_numbers.md). The paper rounds to \$73.

C. JSONL Trace Schema

Each row in results/<run_id>/<cell>.jsonl is one parsed tool call (or one parser-failure record). Schema:

- task_id (str): BFCL-derived task identifier; probe runs append ___<distractor_type> and ___rm when target removed.
- model (str): provider+model id (e.g. claude-sonnet-4-6 or mlx-community/Qwen3-8B-4bit).
- condition (str): one of C0, C0_5, C0_7, C1.
- turn_idx (int): turn number within the task trace.
- parsed_tool_call (object): {name, arguments} or null when the agent emitted no parseable call.
- tool_call_status (str): executed, hallucinated, timed_out, parse_fail, or refused. TEHR numerator counts hallucinated; denominator counts executed + hallucinated.
- tool_response (object), intervention_event (object|null), latency_ms (int), tokens_in (int), tokens_out (int), cost_usd (float), timestamp (str), schema_version (str).

The pass-rate proxy used in TOST is: a task counts as "passing" if its last turn's tool_response contains no error key and no turn in the task has tool_call_status = timed_out. This proxy is documented in scripts/aggregate_all.py and is the surface used to compute Sonnet/Haiku $60/60 \to 53/60, 56/60$.

D. Cross-Family Replication Plan (Future Work)

The harness ships parser support for the Anthropic and OpenAI tool-use schemas plus the MLX adapter;

Llama-3.1's <|python_tag|> response format and Mistral's tokenizer- specific tool-call envelopes are not yet parsed. We document the target replication panel here as future work that the released harness is structurally ready to support: Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Gemma-2-9B-it, Qwen2.5-7B-Instruct, Phi-3.5-mini-instruct, and DeepSeek-R1-Distill-Qwen-7B (all available as 4-bit MLX variants on mlx-community/). Purpose: test whether the matched_random/synonym signature observed in the Qwen3 family at 4B–14B replicates outside Qwen3. The cross-family dimension is therefore explicitly outside this submission's empirical envelope (see §??).

E. Other Limitations and Reproducibility Details

Failure-mode taxonomy. TEHR measures registry-membership violations only. Adjacent failure modes are out of scope: namespace-prefix hallucination (internal.get_user when the registry contains get_user); argument-signature hallucination (correct tool name, fabricated argument types or values); return-shape hallucination (correct call, incorrect downstream expectation). A multimodal extension is straightforward since the failure is upstream of modality, but is not measured here.

Registry-size scaling. RVR's per-call overhead is bounded by the rendered size of $\mathcal{R}$. In our regime ($\sim$5–25 tools) the re-prompt adds <5 tokens. At enterprise-scale registries (100–500+ tools) the registry list scales linearly in $|\mathcal{R}|$ and may dominate the token budget at the tail; hierarchical or retrieval-conditioned variants of RVR (semantic-relevance-filtered registry echo) would mitigate this, but we do not evaluate them here.

Reproducibility. Local-tier MLX inference on Apple Silicon is not bitwise-deterministic across runs due to non-deterministic Metal kernels; per-task tool-call decisions are stable to within ± 1 token, not affecting TEHR aggregates. API-tier decoding is locked at temperature $= 0$ where supported (Sonnet/Haiku 4.x); Opus 4.7 has deprecated the temperature parameter and we use the provider's default. Rolling-alias model strings are resolved at run-start via each provider's models.list endpoint and logged to the reproducibility manifest. Per-task wall-clock cap 120 s and turn cap 8.

F. Per-Cell Upper-Bound Tables

A machine-readable per-cell table (numerator, denominator, Clopper–Pearson lower/upper 95% bounds) is included in the supplementary release as PHASE0/RESULTS/stats_report.json. Selected rows:

- Anthropic 4.x pooled (regime + probe + RVR): C_0 alone 0/2,599, CP $\leq 0.115\%$; C_1 alone 0/678, CP $\leq 0.441\%$.
- Qwen3-1.7B C0: 2/210 = 0.95%; CP 95% upper bound 2.97%.
- Qwen3-4B C0: 3/226 = 1.33%; CP 95% upper bound 3.40%.
- Qwen3-8B C0: 4/269 = 1.49%; CP 95% upper bound 3.37%.
- Qwen3-14B C0: 5/268 = 1.87%; CP 95% upper bound 3.88%.
- Qwen3-32B C0: 0/224 = 0%; CP 95% upper bound 1.33%.